

University of St. Gallen

School of Management, Economics, Law, Social Sciences, International
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Stochastic Methods in Finance

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Table ?? reports descriptive statistics of the payoff $A_T = \max(S_n - \bar{S}_n, 0)$ computed at the terminal nodes of the binomial tree ($t = 25$).

Statistic	Value (\$)
Count	3.36×10^7
Mean	15.71
Std	24.71
Min	0.00
25%	0.00
Median	0.00
75%	24.65
Max	321.58

Table 1: Descriptive statistics of the payoff A_T at the terminal nodes ($t = 25$).

The distribution of A_T is highly right-skewed: the median is zero, meaning more than half of all terminal paths yield a worthless option, i.e. $S_n < \bar{S}_n$. Notably, the standard deviation (24.71) exceeds the mean (15.71), reflecting the high dispersion of payoffs. Most paths expire worthless while a small number of highly favourable paths drive the mean upward. The maximum payoff of 321.58 corresponds to the path where the stock price increases at every period.

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